21 · The retina: rods, cones and colour blindness

The words of this track, as it says them. Bold brown is the examiner.

The retina: rods, cones, and colour blindness.

What receptors are impacted by colour blindness? And what are the receptors for light, where exactly are they found, and how do they differ?

Her checklist puts it like this: the physiology of vision includes how the retina works, and colour blindness is related to the function of the photoreceptors. Her student-marked quiz asks the colour-blindness part straight out.

The hook: cones for colour and detail, in the centre. Rods for dim light, round the edges.

The receptors for light are photoreceptors: rods and cones, in the retina, the inner layer of the eye. They contain visual pigments, and they convert light energy into nerve impulses.

Cones sit in the macula lutea, packed most tightly in the fovea centralis, the very centre of your visual field. They respond to bright light, which is photopic vision. They give colour vision, and they form the sharpest images. Her key: focusing an object on the fovea centralis gives the highest visual acuity.

Rods are more numerous. They're absent from the fovea and the macula, and there are more of them towards the edges of the retina. They respond to dim light, which is scotopic vision, and they see black, white and grey, with no colour. Why are rods sensitive but blurry? Her key: as many as a hundred rods can connect to a single ganglion cell. Lots of signal, not much detail.

A few one-mark details from her quizzes. Visual pigments are told apart by their opsin protein. Light makes the photoreceptor hyperpolarise, and that's what's ultimately signalled to the brain. The part of the retina with no photoreceptors at all is the optic disc, the blind spot, not the macula: her quiz offers the macula as the wrong answer. And an afterimage is when you can still see something after the stimulus has gone, because the retina's signal persists.

Now her model answer for colour blindness, word for word. The receptors impacted are photoreceptors, specifically the cones of the eye. The cones are found mostly in the macula lutea, the central part of the retina that is the centre of your visual field, and the fovea centralis, the very centre of your visual field, where your vision is the clearest. In red-green colour blindness, it's the L cones, for red, and the M cones, for green. And it's X-linked recessive, which is why it's more common in males.

Once more. The retina, and colour blindness.

The photoreceptors are rods and cones, in the retina. Cones sit in the macula lutea, packed in the fovea centralis: bright light, colour, and the sharpest vision. Rods are absent from the fovea and increase towards the edges: dim light, black and white, and many rods share one

ganglion cell. Pigments differ by their opsin, and the optic disc is the blind spot. Colour blindness affects the cones, found mostly in the macula lutea and the fovea centralis: in red-green blindness, the L and M cones.